



FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☒ Mandatory on Request	GEMS - A:	Close Date:

- ☐ **Staged FMI**
 - ☐ **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
 - ☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
 - ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
 - ☐ **No Parts** are required to complete this FMI.
- ☐ **Batch with**

To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25353

FMI Title: LS 2X/3X/4X, HS QXi - Software Patch & Tube ID Change

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

This FMI is an optional field upgrade kit requested by the customer for systems (with indicated gantry types) that have Irix/Pegasus or Linux/Pegasus systems.

- LightSpeed 2X: 2260619 and 2260619-3
- LightSpeed 3X: 228117, 228117-2, 228117-3 and 2281177-4
- LightSpeed 4X: 2339985
- HiSpeed QX/I: 2341799

Not applicable for systems with GRE consoles

Purpose

The purpose of this FMI is to load Patch F.2 on IRIX/Pegasus Consoles running LS2002M5 software or Patch B.2 on Linux/Pegasus – GOC2 consoles running M4 software.

Also to add an adaptor cable to change the tube ID on systems specified so that the software can identify whether the system uses Performix Ultra tubes.

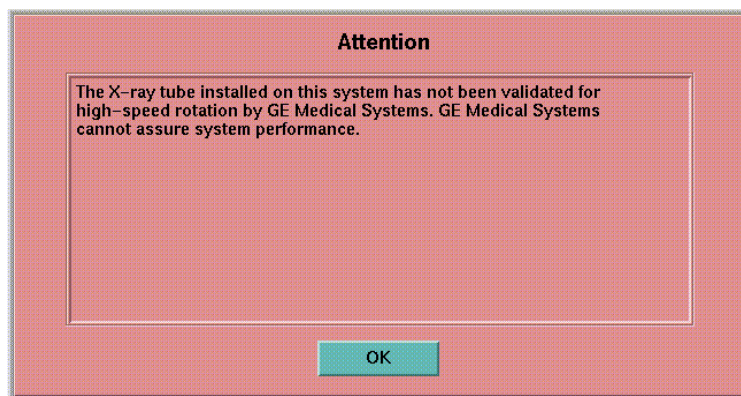
IMPROVEMENTS

LS2002M5 Patch F.2

New Features

- Smartube

Software and Hardware have been added to assure that a GE Medical Systems validated x-ray tube for the system configuration is installed. If an un-validated tube is found, the following warning dialog will be posted at system start up.



Product Safety Fixes

- Scout Image Fails to recon.

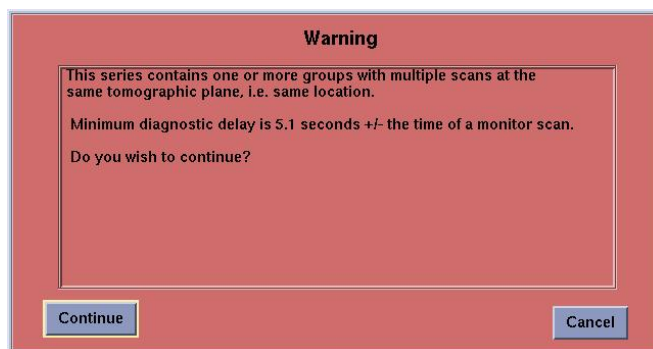
This was caused by characters added at the end of the Exam Description when the shift or control key was pressed along with the enter key.

- Additional slices added when scans are prescribed with gantry tilt.
Improvements have been made in Graphic Rx and ScanRx.
- Show Localizer lines are in wrong location if user skips series quickly.
- DLP is higher than expected caused by selecting Repeat Series Quickly.
The dose reported was incorrect.

Improvements

- SmartPrep Diagnostic delay not as expected.

The diagnostic delay is adjusted at Confirm to account for table movement required between the Monitor and Scan phase locations. If the Diagnostic Delay prescribed is less than what the system can do, the expected delay will be posted at Confirm.



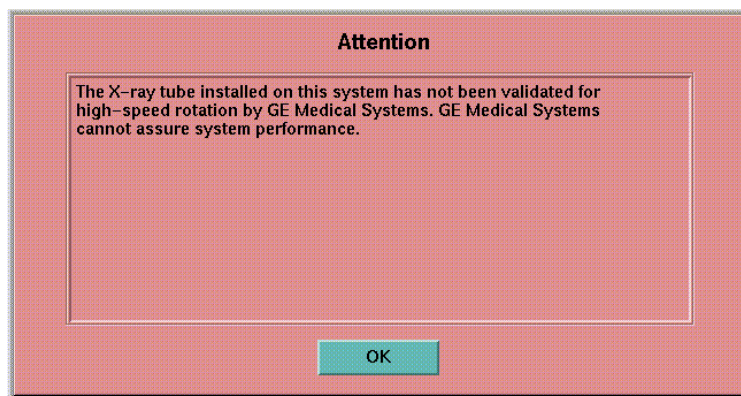
- SmartPrep fails to start.
- Improvements have been made in communication between the ScanRx and Display Processes to assure reliable use of SmartPrep
- Start Locations have odd numbers.
Adjustments to Start Locations from Graphic Rx will be rounded off in .25mm increments. This will give more meaningful scan locations for the user.
- Images can't be retrieved from PACS.
- System shuts down with a ScanRx crash.
- Dcserver crashes with Storage Commitment requests.
- Images are filmed cut off.
This was caused when filming images with GSE from a non-square viewport.
- Update Apache HTTP server to be more secure.
- Clean up Pro Diags so Reli Monitor does not fail.

GOC2 Linux/Pegasus Patch B.2

New Features

- Smartube

Software and Hardware have been added to assure that a GE Medical Systems validated x-ray tube for the system configuration is installed. If a un-validated tube is found, the following warning dialog will be posted at system start up.



Product Safety Fixes

- Scout Image Fails to recon.

This was caused by characters added at the end of the Exam Description when the shift or control key was pressed along with the enter key.

- Additional slices added when scans are prescribed with gantry tilt

Improvements have been made in Graphic Rx and ScanRx.

- Show Localizer lines are in wrong location if user skips series quickly.
- Smart Step Images out of sequence.
- Auto mA changes automatically or reduces to 10 mA.

This was due to how Noise index was updated when the user changed slice thickness. Prior to this patch the Noise index adjustments were made from the Reference Noise Index. The change is to adjust Noise Index on Slice Thickness change with the Rx'd Noise Index. This will assure that when the users changes slice thickness and then returns to the original slice thickness the Noise Index will default to the original value.

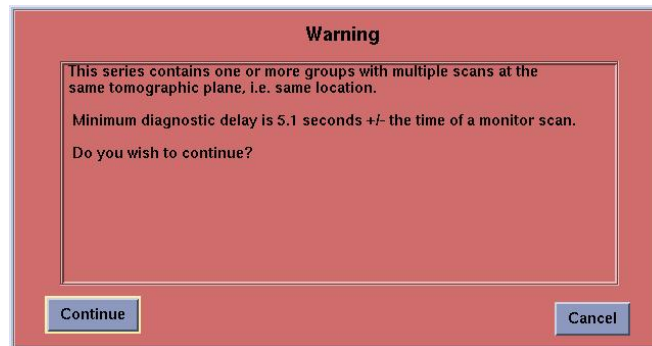
- System recommends optimize to 35 mA.
- Show Localizer lines are in wrong location if user skips series quickly.
- DLP is higher than expected caused by selecting Repeat Series Quickly.

The dose reported was incorrect.

Improvements

- Head recon is slow.
- Apostrophe in Hospital Name causes Tube Warm Up to fail.
- SmartPrep Diagnostic delay not as expected.

The diagnostic delay is adjusted at Confirm to account for table movement required between the Monitor and Scan phase locations. If the Diagnostic Delay prescribed is less than what the system can do, the expected delay will be posted at Confirm.

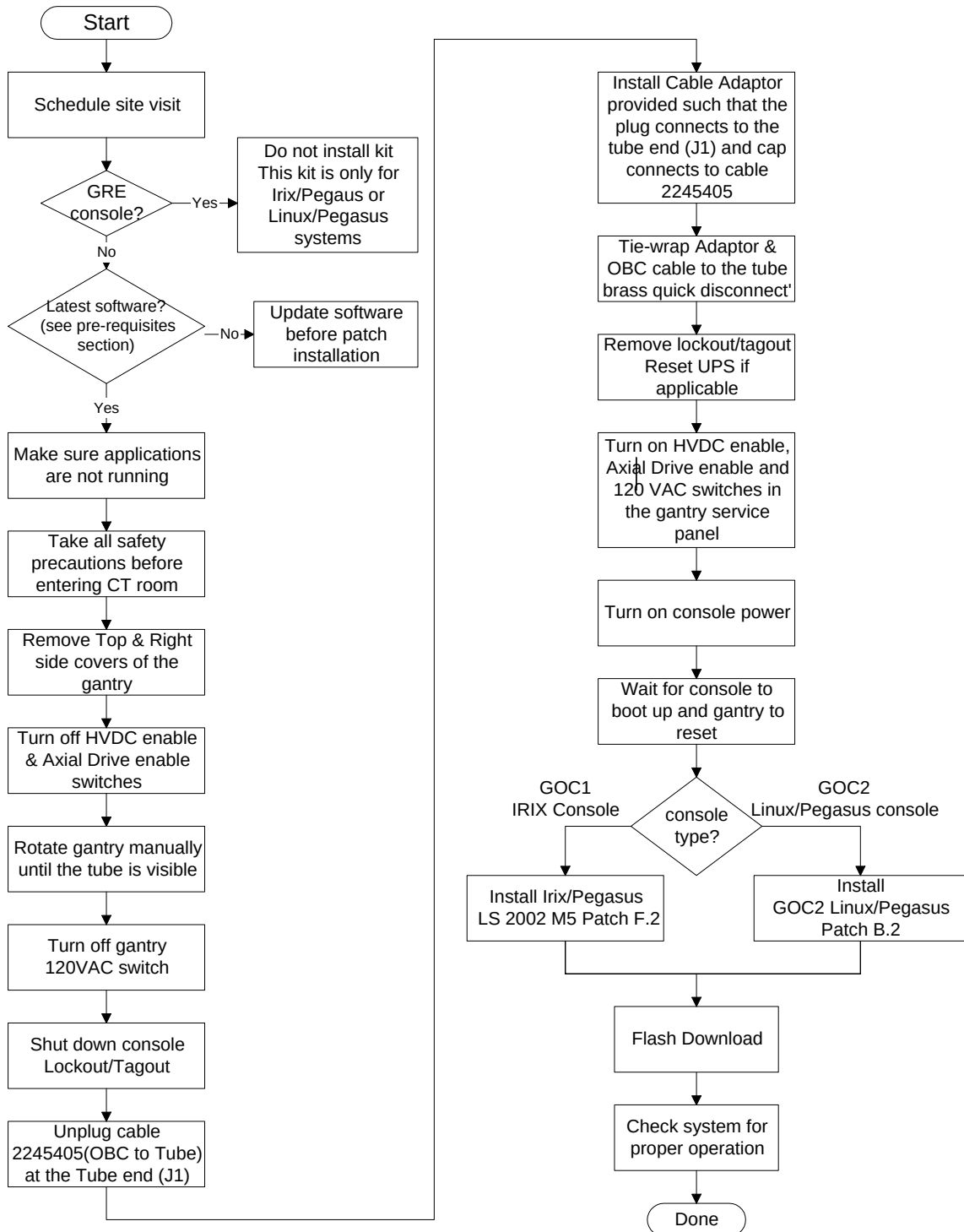


- SmartPrep fails to start.
- Improvements have been made in communication between the ScanRx and Display Processes to assure reliable use of SmartPrep.
 - Scan aborts due to Axial not ready.
- Start Locations have odd numbers.

Adjustments to Start Locations from Graphic Rx will be rounded off in .25mm increments. This will give more meaningful scan locations for the user.

 - Display hung with 10% loading
 - System Hung with Can not dismiss Attention Window
- Images can't be retrieved from PACS.
- System shuts down with a ScanRx crash.
- Dcserver crashes with Storage Commitment requests.
- Auto Voice fails to play if the table is moved during a long prep delay.
- Images are filmed cut off.

This was caused when filming images with GSE from a non-square viewport.
- Update Apache HTTP server to be more secure.
- Clean up Pro Diags so Reli Monitor does not fail.

Process Flow

Procedure

1 Preliminary Checks

Check to make sure that the system has the latest software. The latest software versions are listed in the Prerequisites section of this document. If the system does not have the latest software, please update the software before installing this FMI

Notice:

- ***Do not install this FMI to systems with GRE Consoles***
- ***Do not install this FMI to CT/PET systems***
- ***As it is not necessary to remove the tube for this FMI, please do not replace tube prematurely as it will have negative impact on tube availability***

2 Installation of the Cable Adaptor, 5127930

1. Make sure that applications are not running.
2. Take all safety precautions before entering the scan room.
3. Lower table to home (lowest) position.
4. Remove gantry right top cover and right side cover.

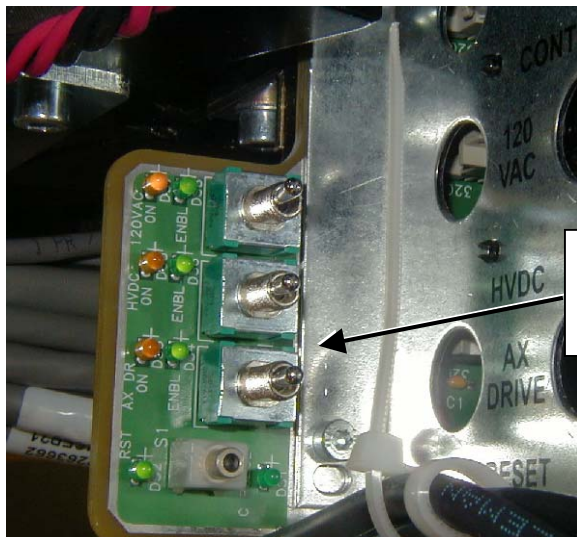


WARNING



FAILURE TO DISABLE THE AXIAL DRIVE MAY RESULT IN SERIOUS INJURY DUE TO UNEXPECTED GANTRY ROTATION.

5. Turn off high voltage DC enable switch (HVDC) and the Axial Drive enable (AX DR) switch to off position in the service panel inside the gantry.



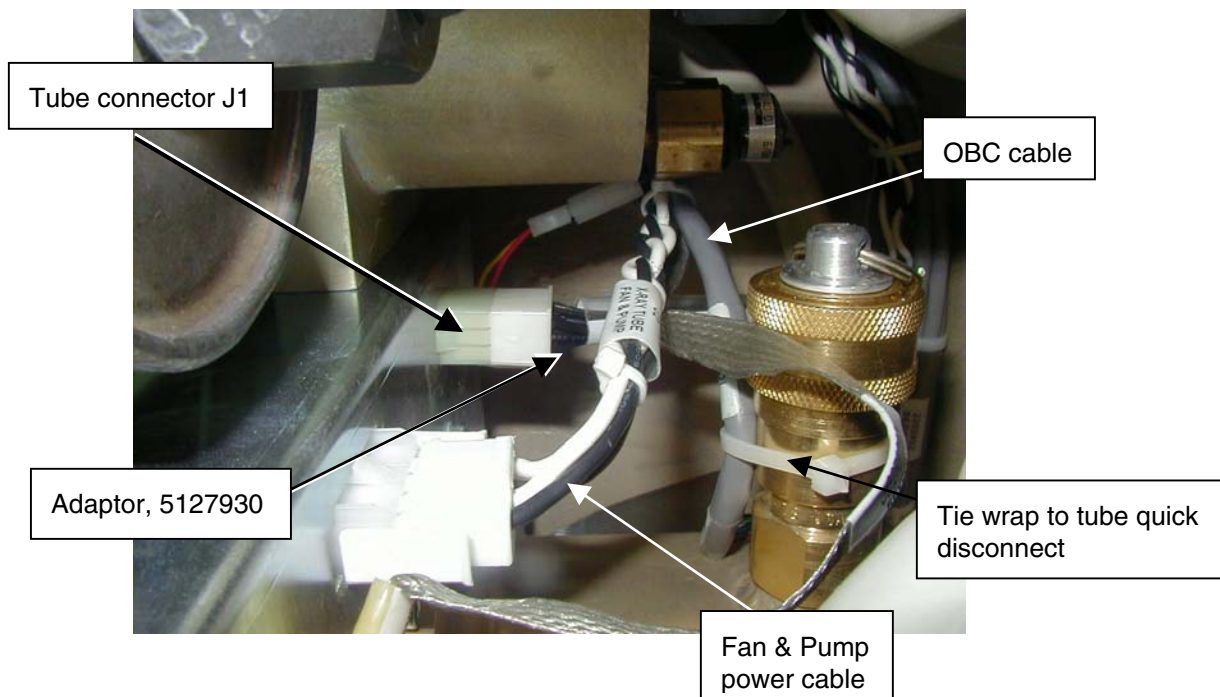
HVDC enable,
Axial Drive enable and
120VAC switches

6. Rotate the gantry with care until the tube is visible.
7. Now, turn off the 120VAC switch in the service panel
8. Shut off power to the console and follow lockout/tagout procedure to make sure that power is off while making the changes to the gantry.
9. ***Install Adaptor, 5127930, only on systems with GE Performix Ultra tube. This can be verified by checking the tube insert model number (2120785-2), which can be found on tube rating plate. Do not install adaptor for any other tube. For other tubes, move to 'Installation of Software Patch' section (section 3).***

Note:

For any subsequent tube changes, the adaptor should remain with the gantry.

10. Locate cable 2245405, which connects OBC to the Tube ID connector J1. It is behind the tube cooling tank power cable.
11. Unplug the cable 2245405 **only at the tube end (at J1)**.
12. For easy access to tube connector J1, if you have to disconnect Fan & Pump power cable, ***REMEMBER TO RECONNECT AT COMPLETION.***
13. Insert the cable Adaptor (5127930) supplied with this FMI, between the OBC cable 2245405 and tube ID connector J1 such that plug of the Adaptor connects to the cable and the cap end plugs to the tube connector J1.
14. Tie-wrap Adaptor and the OBC cable to the tube 'brass quick disconnect'.



15. IMPORTANT: Visually check the gantry to make sure that:
 - ***The Adaptor is connected correctly and tie-wrapped***
 - ***The power cable to the tube cooling tank is connected***
 - ***The tube ground wire is in place***
16. Remove lockout/tagout so that power can be applied to the system. If the site uses UPS, reset UPS.
17. Turn on all three switches (Axial Drive, HVDC and 120VAC) in the gantry.
18. Power up console.
19. Wait for the console to boot up and the gantry to reset.

3 Installation of Software Patch

If cable adaptor is needed (see Installation of cable adaptor section), make sure that the adaptor is installed before loading the patch.

3.1 IRIX/PEGASUS Systems

Patch Installation:

You need the CDROM LS2002 M5 Patch F.2.

1. Shut down all CT Applications.
2. Open a Unix Shell.
3. Type **su** – <Enter>
4. Type password: **#bigguy**
5. Insert LS2002 M5 Patch F.2 CD into the CT CD drive.
6. Type **mountCD /mnt** <Enter>
7. Type **cd /mnt** <Enter>
8. Type **./install_patch -p ls2002_m5_patch_F_2** <Enter>
9. Type **cd /** <Enter>
10. Type **umount /mnt** <Enter>
11. Type **eject 1 6** <Enter>
12. Remove the CD out of the disk drive.
13. Type **reboot** <Enter>
14. Once the system and applications have started up, open a UNIX shell, and as ct user type: **index_img_uid** <Enter>
15. Select Service Desk Top.
 - Select Utilities.
 - Select Install Folder.
 - Select Flash Down Load Tool.
 - Select Update.
 - Dismiss after done.

16. Select Pink Shutdown button.

Select Restart.

17. Open a Unix Shell.

18. Confirm that the patch has been successfully installed by typing:

showprods | grep ls2002_M5_patch_F_2 <Enter>

The output will be as follows:

```
I ls2002_M5_patch_F_2 02/05/2005 Patch F.2 for ls2002 M5
I ls2002_M5_patch_F_2.img 02/05/2005 Patch files for ls2002 m5
I ls2002_M5_patch_F_2.img.ocroot 02/05/2005 Files that go onto ocroot.
```

19. Clean up of idm.LOG file:

Type **cd /usr/g/insite/idm** <Enter>

Type **tail -500 idm.LOG > idm.TMP** <Enter>

Type **cp -f idm.TMP idm.LOG** <Enter>

When message “/bin/cp: overwrite ‘idm.LOG’ ?” comes up, type **y** <Enter>

Type **rm idm.TMP**

When message “/bin/rm: remove ‘idm.TMP’ ?” comes up, type **y** <Enter>

20. If there is a need to un-install, see Appendix A for instructions.

3.2 Linux/Pegasus Systems

You need the CDROM Linux Pegasus M4 Patch B.2.

Patch Installation:

1. Shut down all CT Applications.
2. Insert Linux Pegasus M4 Patch B.2 CD into the CD drive.
3. Open a Unix Shell.
4. Type **su -** <Enter>
5. Type password: **#bigguy**
6. Type **mount /mnt/cdrom** <Enter>
7. Type **cd /mnt/cdrom** <Enter>
8. Type **./install_patch -p Lnx_Peg_M4_Patch_B_2-1-1.i386.rpm** <Enter>
9. Type **cd /** <Enter>
10. Type **umount /mnt/cdrom** <Enter>
11. Type **eject** <Enter>
12. Remove the CD from the disk drive.
13. Type **reboot** <Enter>

14. Select Service Desk Top.
 - Select Utilities.
 - Select Install Folder.
 - Select Flash Down Load Tool.
 - Select update.
15. Select Pink Shutdown button.
 - Select Restart.
16. Open a Unix Shell.
17. Confirm that the patch has been successfully installed by typing:
showprods | grep Lnx_Peg_M4_Patch_B_2<Enter>

The output will be as follows:

```
I Lnx_Peg_M4_Patch_B_2    02/05/2005    Linux Pegasus M4 Patch B.2 ver 1 rel 1_1
```

18. Clean up of idm.LOG file:
 - Type **cd /usr/glinsite/idm** <Enter>
 - Type **tail -500 idm.LOG > idm.TMP** <Enter>
 - Type **cp -f idm.TMP idm.LOG** <Enter>
 - When message “/bin/cp: overwrite ‘idm.LOG’ ?” comes up, type **y** <Enter>
 - Type **rm idm.TMP**
 - When message “/bin/rm: remove ‘idm.TMP’ ?” comes up, type **y** <Enter>
19. If there is a need to un-install, see Appendix B for instructions.

4 System Checkout

Perform the following scans (Abdominal or Head):

50 images at 0.5 sec rotation at 120KV, 40mA

The system should scan and image without any errors.

5 Finishing Up

1. Make sure that the ‘Tips and Workaround’ document supplied with this FMI kit is left with the customer. 5115418-100 for IRIX systems and 2370773-100 for Linux/Pegasus systems. Also, leave the Customer Letter, 5128656-100 with the customer.
2. Close FMI per regional procedure. Note that the FMI cannot be closed until the Product Locator cards are completed, sent in, and GIB is updated with all tracked items in this FMI.
3. **Remove this FMI document from the customer site. If the cable Adaptor is not used, please discard. Do not leave it at the customer site.**

Appendix A: IRIX/Pegasus Systems – Patch Un-Installation

1. Shutdown all CT applications.
2. Insert Put the LS2002 M5 Patch F.2 CD into the CD drive.
3. Open a Unix shell.
4. Type **su** - <Enter>
5. Type password: **#bigguy**
6. Type **mountCD /mnt** <Enter>
7. Type **cd /mnt** <Enter>
8. Type **./uninstall_patch** <Enter>

This will display a menu with all the available patches installed. In this case only LS2002 M5 Patch F.2 will be shown.

The installed patches are listed below.

Please selected the number associated with the patch to be uninstalled.

Select 0 to exit.

0 - exit

1 - ls2002_M5_patch_F_2

Enter a number:

9. Type **1** <Enter>
10. Type **cd /** <Enter>
11. Type **umount /mnt** <Enter>
12. Type **eject 1 6** <Enter>
13. Remove CD from the CD drive.
14. Type **reboot** <Enter>
15. Once the system has rebooted run Flash Download (see step 15 of the install Instructions).
16. Select Pink Shutdown button.
Select Restart.

Appendix B: Linux/Pegasus Systems – Patch Un-Installation

1. Shutdown all CT applications.
2. Insert Linux Pegasus M4 Patch B.2 CD into the CD drive.
3. Open a Unix shell.
4. Type **su** – <Enter>
5. Type password: **#bigguy**
6. Type **mount /mnt/cdrom** <Enter>
7. Type **cd /mnt/cdrom** <Enter>
8. Type **./uninstall_patch** <Enter>

This will display a menu with all the available patches installed. In this case only Linux Pegasus M4 Patch B.2 will be shown.

The installed patches are listed below.

Please selected the number associated with the patch to be uninstalled.

Select 0 to exit.

0 - exit

1 - Lnx_Peg_M4_Patch_B_2-1-1

Enter a number:

9. Type **1** <Enter>
10. Type **cd /** <Enter>
11. Type **umount /mnt/cdrom** <Enter>
12. Type **eject** <Enter>
13. Remove the CD from the disk drive.
14. Type **reboot** <Enter>
15. Once the system has rebooted run Flash Download (see step 14 of the install Instructions).
16. Select Pink Shutdown button.

Select Restart.

NOTES: